

Market-Pulse-NN: Real-Time Market Movement Prediction

Using Sequential Deep Learning

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Abstract—Predicting short-term equity market direction is a challenging problem that requires fusing heterogeneous information streams — price dynamics, technical indicators, and investor sentiment — across time. This paper presents *Market-Pulse-NN*, an end-to-end pipeline that ingests price data from Yahoo Finance, pre-labelled historical news sentiment from the Alpha Vantage News Sentiment API, recent free-text headlines from NewsData.io, and Reuters RSS feeds, derives per-document sentiment scores using FinBERT (with a VADER fallback) for the free-text sources, and constructs fixed-length multivariate sequences for next-day binary direction classification. We train and rigorously compare three sequential deep learning architectures — Vanilla Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) — all sharing an identical training regime to isolate the effect of architectural choice. Evaluation on a chronologically held-out test split across five large-cap U.S. equities (AAPL, TSLA, NVDA, MSFT, AMZN) reveals a striking pattern: both Vanilla RNN and LSTM collapsed to a single-class prediction policy on the test set (yielding identical metrics of 44.66% accuracy and 0.31 macro-F1), whereas GRU produced the only non-degenerate decision boundary, achieving 54.17% accuracy and 0.53 macro-F1. ROC-AUC values for all three models cluster narrowly in the 0.52 range, confirming the well-documented difficulty of next-day market direction prediction in near-efficient markets. We discuss the role of severe sentiment data sparsity (only 6.5%–10.8% of trading days carried real sentiment signal under free-tier API constraints) as a likely contributor to the model-collapse phenomenon. All code, model weights, and reproducible notebooks are released publicly.

Index Terms—sequential deep learning, market prediction, sentiment analysis, LSTM, GRU, recurrent neural network, FinBERT, Alpha Vantage, time-series classification

I. INTRODUCTION

Financial market prediction has long attracted both academic and practitioner interest. Traditional quantitative approaches rely on technical indicators derived from historical price and volume data [3]. While effective as inputs, they do not capture the rich, unstructured information encoded in news articles, earnings calls, and analyst commentary — signals that can move prices hours or days before they appear in price charts.

The proliferation of financial news APIs now makes it feasible to ingest this unstructured data at scale. Simultaneously, pre-trained transformer-based language models such as FinBERT [1] provide high-quality, domain-adapted sentiment

classification without requiring task-specific labelled corpora. Combined with sequential deep learning architectures capable of modelling temporal dependencies, these advances open the door to sentiment-augmented time-series models for market prediction.

A. Motivation

The core hypothesis of this work is that *fused* signals — price dynamics, technical indicators, and aggregated sentiment — carry more predictive information about next-day direction than price features alone. We further hypothesised that gated architectures (LSTM, GRU), which mitigate the vanishing-gradient problem, would outperform the Vanilla RNN on the 60-day lookback windows used in our experiments. As Section V will show, the empirical reality is more nuanced than this hypothesis suggested.

B. Contributions

The key contributions of this paper are:

- A fully automated, modular data-ingestion pipeline spanning four complementary sources: Yahoo Finance price data, Alpha Vantage pre-labelled historical news sentiment, NewsData.io recent financial headlines, and Reuters RSS feeds.
- A unified sentiment scoring interface combining FinBERT (GPU) and VADER (CPU fallback) for free-text sources, integrated with Alpha Vantage’s native Bullish/Neutral/Bearish labels and ticker-relevance scores.
- Comprehensive architectural comparison of RNN, LSTM, and GRU under identical training conditions on five tickers, with an honest analysis of model-collapse behaviour observed in two of the three architectures.
- Publicly available code, pre-processed datasets, all generated figures, and the present Overleaf report source.

The remainder of this paper is structured as follows. Section II describes the full pipeline methodology. Section III presents dataset statistics and coverage analysis. Section IV details the model architectures. Section V reports and interprets the experimental results. Section VI discusses challenges encountered. Section VII concludes the paper.

II. METHODOLOGY

Figure 1 shows the overall system pipeline. The six stages are described below.

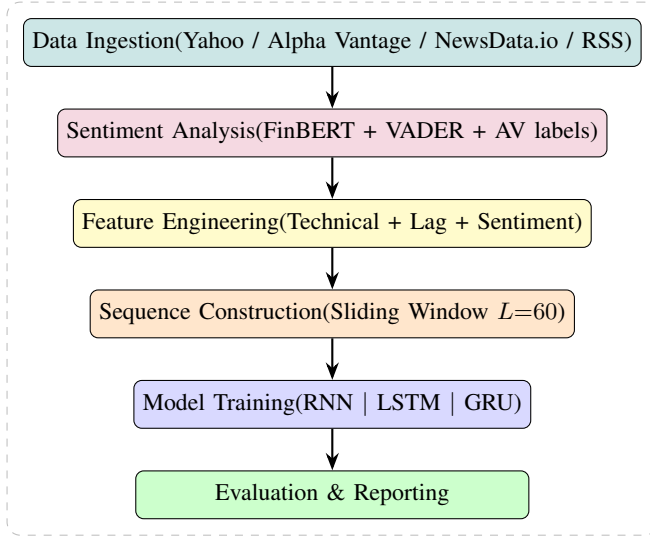


Fig. 1: End-to-end Market-Pulse-NN pipeline.

A. Data Ingestion

Four complementary data sources are ingested:

- **Yahoo Finance** (`yfinance`): Daily OHLCV price bars for five tickers spanning January 2021 through December 2024 (≈ 1008 trading days per ticker).
- **Alpha Vantage News Sentiment API**: Pre-labelled historical financial news articles, each returned with a Bullish/Neutral/Bearish overall label, a numeric sentiment score in $[-1, +1]$, and a per-ticker relevance score in $[0, 1]$. The free tier limits the number of calls per day and returns historical articles clustered around discrete date windows.
- **NewsData.io**: Recent free-text financial headlines queried via the `$AAPL`, `$TSLA`, `$NVDA`, `$MSFT`, `$AMZN` cashtag filters. The free tier restricts queries to a recent rolling window, limiting historical coverage.
- **Reuters RSS** (`feedparser`): Four feeds (`reuters_rssindex`, `reuters_topstories`, `reuters_news_25`, `reuters_mw_realtimeheadlines`) polled at ingestion time, returning current and very recent headlines.

All text artefacts are stored as JSON alongside their UTC publication timestamp for later time-alignment with price data. Figure 3 summarises per-source temporal coverage.

B. Sentiment Analysis

Each text document is mapped to a numeric sentiment score in $[-1, +1]$ and a categorical label in $\{positive, negative, neutral\}$.

Alpha Vantage articles arrive pre-scored. We use the provided `ticker_sentiment_score` field directly as the

numeric sentiment, with Bullish $\rightarrow$ positive, Bearish $\rightarrow$ negative, Somewhat-Bullish/Somewhat-Bearish/Neutral mapped to positive/negative/neutral by threshold.

NewsData.io and Reuters RSS articles are raw free-text and are scored locally:

- **Primary** — FinBERT [1], a BERT-based model fine-tuned on the Financial PhraseBank corpus. Inference runs in batches of 64 with a maximum token length of 512. On Google Colab the model is moved to an NVIDIA A100 GPU; locally on macOS it runs on CPU.
- **Fallback** — VADER [2], a lexicon-and-rule based sentiment engine requiring no model download. It is invoked when FinBERT is unavailable (e.g. first local run prior to weight download) and uses a compound score threshold of ± 0.05 to assign labels.

Per-document numeric scores are aggregated into per-ticker, per-trading-day features by joining the document publication timestamp to the next available trading day's close. Aggregation statistics include daily mean sentiment (`av_sentiment_mean`), rolling 3-day and 5-day means (`sent_roll_3d`, `sent_roll_5d`), one- to three-day lags (`sent_lag_{1,2,3}`), and sentiment momentum.

C. Feature Engineering

Each daily observation $\mathbf{x}_t$ is a vector comprising:

- Technical indicators**: RSI (period 14), MACD line and signal (12/26/9), MACD histogram, Bollinger Bands (upper, lower, %B; period 20, 2σ), and Average True Range (period 14), all computed via the `ta` library. Figure 9 illustrates these indicators on NVDA for visual inspection.
- Lagged price features**: Close-price lags $\{c_{t-1}, c_{t-2}, c_{t-3}, c_{t-5}\}$, raw daily return, log return, and volume change.
- Sentiment features**: Daily mean sentiment score, rolling 3-day and 5-day means, lagged sentiment $\{s_{t-1}, s_{t-2}, s_{t-3}\}$, and sentiment momentum (first difference of the daily score).

All features are standardised using a `StandardScaler` fitted *only* on the training split to prevent data leakage. Figure 8 shows the resulting feature correlation matrix for AAPL.

D. Sequence Construction

A sliding window of length $L = 60$ trading days and stride 1 is applied to produce input tensors $\mathbf{X} \in \mathbb{R}^{N \times 60 \times F}$, where N is the number of samples and F is the feature dimension. The target $y_t \in \{0, 1\}$ is the next-day binary direction label ($y_t = 1$ if `closet+1` > `closet`, else 0).

Data are split *chronologically* (no shuffling before the split) to prevent look-ahead bias, using a roughly 70/15/15 train/validation/test ratio per ticker; sequences from all five tickers are then concatenated. Only the training split is shuffled during mini-batch construction.

E. Training Regime

All three models share an identical training configuration to ensure fair architectural comparison:

- **Optimiser:** AdamW with weight decay 10^{-5} , initial learning rate 10^{-3} .
- **Loss:** Binary Cross-Entropy with logits (BCEWithLogitsLoss).
- **Scheduler:** Cosine annealing over 100 epochs.
- **Early stopping:** Patience of 15 epochs, monitored on validation macro-F1.
- **Hardware:** NVIDIA A100 (40 GB) on Google Colab Pro for training; local development and lightweight tasks (data ingestion, evaluation) on a 2020 MacBook running VS Code.
- **Checkpointing:** best-validation-F1 model saved as .pt.

III. DATASET OVERVIEW

A. Tickers and Price Coverage

Five large-cap U.S. equities covering technology and e-commerce sectors were selected: AAPL, TSLA, NVDA, MSFT, and AMZN. The price history spans four calendar years (2021–2024), yielding approximately 1 008 trading days per ticker. Figure 2 shows raw and normalised closing prices over this window, highlighting the dispersion of returns (NVDA appreciated $\approx 10\times$, AMZN $\approx 1.4\times$).

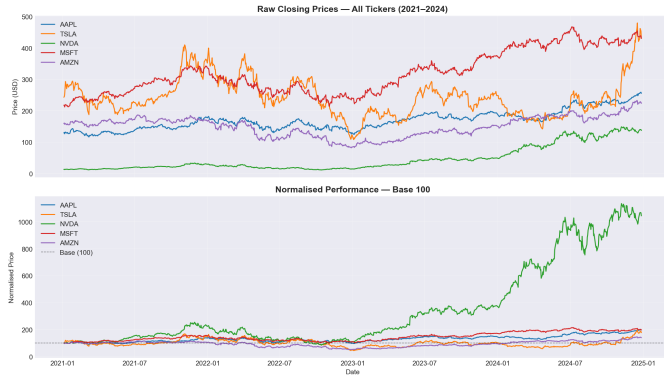


Fig. 2: Closing prices (top) and base-100 normalised performance (bottom) for all five tickers, 2021–2024.

TABLE I: Price Data Summary (2021-01-04 – 2024-12-31)

Ticker	Trading Days	Up Days (%)	Down Days (%)
AAPL	1 008	53.1	46.9
TSLA	1 008	51.9	48.1
NVDA	1 008	53.5	46.5
MSFT	1 008	51.7	48.3
AMZN	1 008	51.3	48.7
Aggregated	5 015	52.3	47.7

The mild class imbalance ($\approx 52\%$ up days) motivates the use of macro-averaged F1 alongside accuracy as the primary evaluation metrics.

B. Sentiment Corpus Statistics

Per-source coverage and document counts are summarised in Table II. Figure 3 visualises weekly record density across all four sources, making clear that Alpha Vantage and Reuters

RSS coverage is concentrated in specific time windows rather than uniform across 2021–2024.

TABLE II: Text Corpus Overview by Source

Source	Window	Scoring	Articles
Alpha Vantage	2021–2024 (sparse)	Pre-labelled	824
NewsData.io	Recent (free tier)	FinBERT + VADER	200
Reuters RSS	2024 onwards	FinBERT + VADER	77
Total			1 101

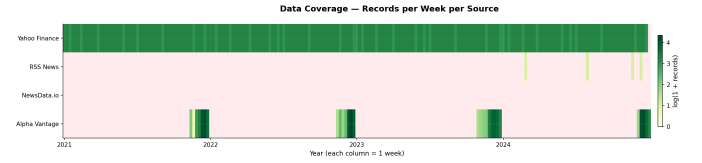


Fig. 3: Per-source record density (log scale) by week, 2021–early 2026. Yahoo Finance provides continuous daily coverage; sentiment sources are intermittent under free-tier API constraints.

C. Alpha Vantage Sentiment Distribution

Alpha Vantage returns 824 articles across the five tickers (Figure 4), with a label distribution of 50.5% Bullish, 36.3% Neutral, and 13.2% Bearish. A high-relevance threshold of 0.5 on the ticker-relevance score retains 811 articles. Mean per-ticker sentiment ranges from +0.041 (TSLA) to +0.237 (MSFT), confirming the documented positive skew of financial news language [4].

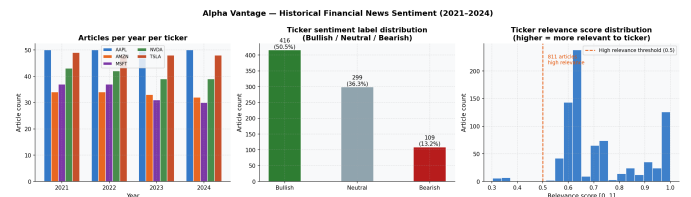


Fig. 4: Alpha Vantage News Sentiment overview: articles per year per ticker (left), label distribution (centre), and ticker-relevance score histogram (right).

D. FinBERT/VADER Scoring of Free-Text Sources

For NewsData.io and Reuters RSS, FinBERT (with VADER fallback) produced the label distributions shown in Figure 5. Both sources skew neutral (RSS 47%, NewsData.io 44%), reflecting the headline-style neutral phrasing typical of newswire reporting.

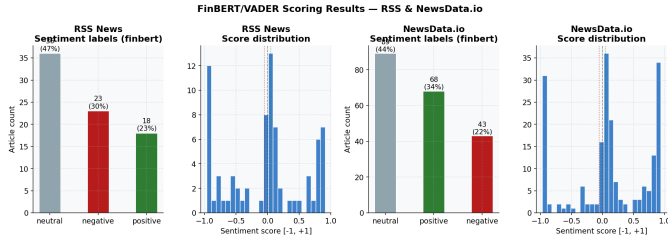


Fig. 5: FinBERT/VADER scoring of RSS and NewsData.io articles: categorical label counts (bars) and numeric score distributions (histograms).

E. Sentiment Density vs. Trading Days

A critical caveat for this work is the *density* of sentiment signal relative to the trading-day calendar. Figure 6 quantifies this: only 6.5% (AMZN) to 10.8% (NVDA) of trading days in our four-year window have any real sentiment observation. The remaining $\approx 90\%$ of days are forward-filled with the most recent observation or imputed as neutral (0). This sparsity is a direct consequence of free-tier API rate limits and historical coverage windows, and likely contributes substantially to the muted performance observed in Section V.



Fig. 6: Daily sentiment score per ticker (top) and percentage of trading days with at least one real article (bottom). Coverage is sparse and clustered, limiting the statistical strength of the sentiment signal.

F. Sequence Dataset

After feature engineering, standardisation, and sliding-window construction, the final dataset comprises 4550 sequences (Figure 7), split chronologically as 3185 train / 682 validation / 683 test. Class balance is broadly consistent across splits (Train 51.9% UP, Val 52.2% UP, Test 55.3% UP), though the test split shows a slight upward shift that will become relevant in Section V.

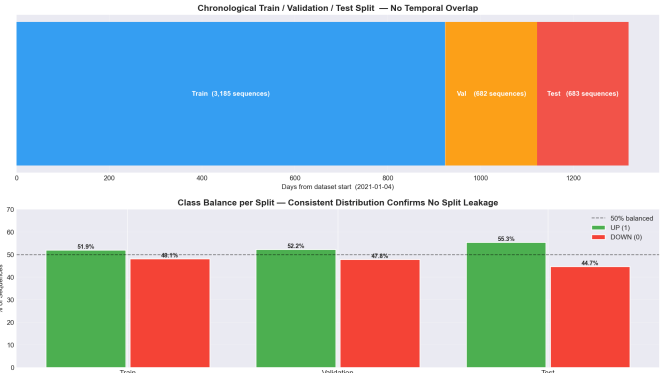


Fig. 7: Chronological train/val/test split (top) with no temporal overlap, and per-split class balance (bottom). The test set exhibits a slightly higher up-day proportion than train or validation.

G. Feature Correlation Structure

Figure 8 shows the Pearson correlation matrix for the AAPL feature set. Sentiment features (`av_sentiment_mean`, `sent_roll_*`, `sent_lag_*`) exhibit modest positive correlations ($r \approx 0.18\text{--}0.25$) with price-level features but near-zero correlation with the daily return target, foreshadowing the limited predictive signal extractable from the available sentiment data.

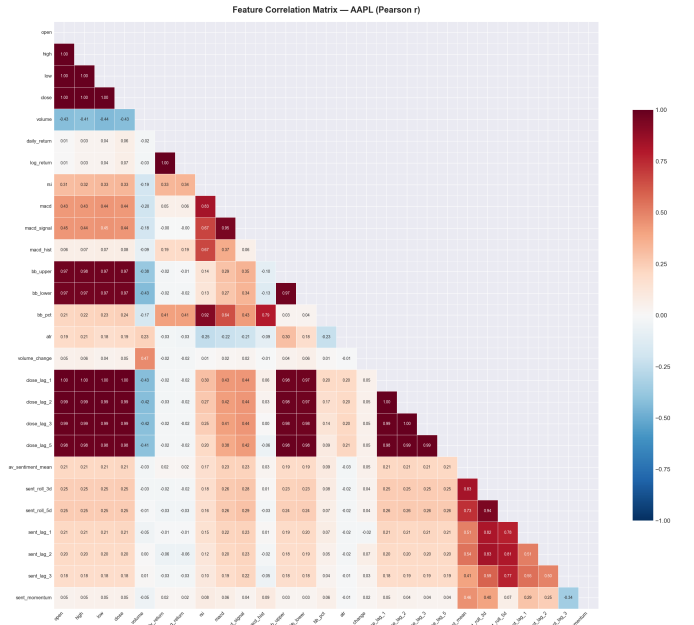


Fig. 8: Pearson correlation matrix of all engineered features for AAPL. Sentiment features (lower-right block) are only weakly correlated with daily returns.

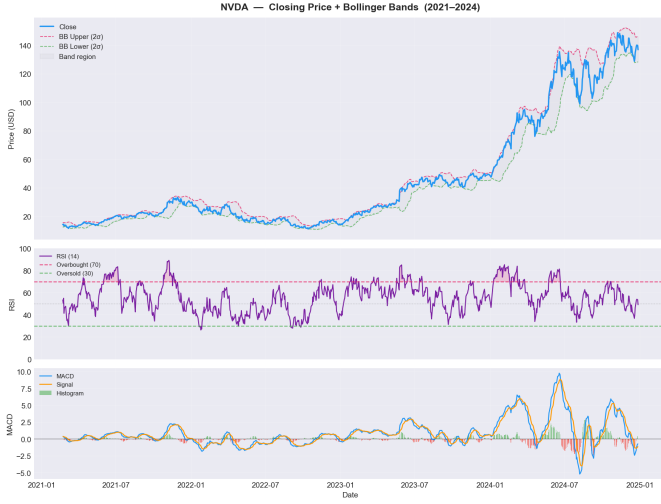


Fig. 9: Visual sanity check of technical indicators on NVDA: closing price with Bollinger Bands, RSI (with 70/30 thresholds), and MACD (line/signal/histogram).

IV. MODEL ARCHITECTURES

All three models map an input sequence $\mathbf{X} \in \mathbb{R}^{B \times 60 \times F}$ to a scalar logit $\hat{y} \in \mathbb{R}^B$, which is passed through a sigmoid to obtain the probability of an up-day. The shared *classification head* consists of a linear layer ($128 \rightarrow 64$), ReLU activation, a 0.3 dropout layer, and a final linear layer ($64 \rightarrow 1$).

A. Vanilla RNN

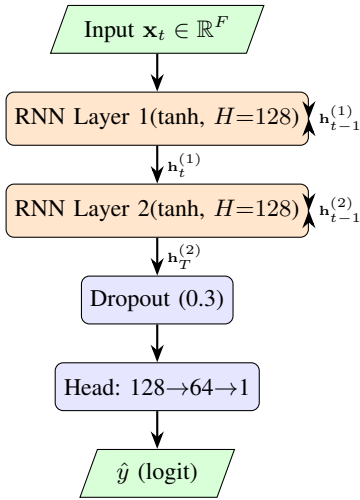


Fig. 10: Vanilla RNN architecture. Only the final hidden state $\mathbf{h}_T^{(2)}$ is fed to the classification head.

The hidden state update rule is:

$$\mathbf{h}_t = \tanh(\mathbf{W}_h \mathbf{h}_{t-1} + \mathbf{W}_x \mathbf{x}_t + \mathbf{b}) \quad (1)$$

The absence of gating mechanisms makes this architecture prone to vanishing gradients over the 60-step horizon.

B. Long Short-Term Memory (LSTM)

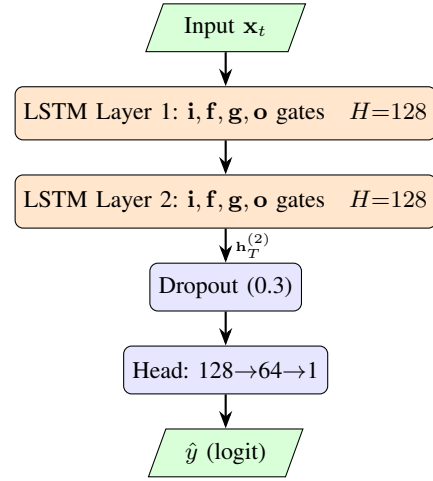


Fig. 11: LSTM architecture. The cell state $\mathbf{c}_t$ enables long-range dependency modelling.

The LSTM cell equations governing forget (f), input (i), cell (c), and output (o) gates are:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f) \quad (2)$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i) \quad (3)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c) \quad (4)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad (5)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o) \quad (6)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (7)$$

C. Gated Recurrent Unit (GRU)

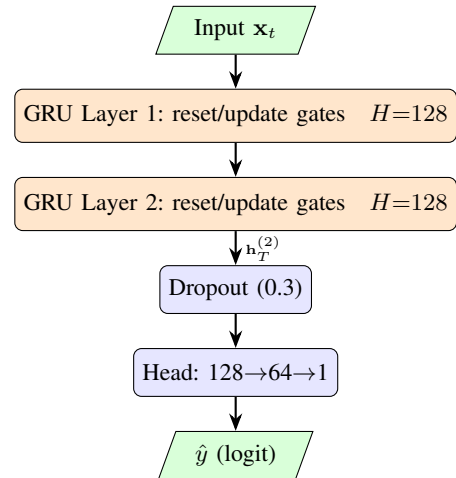


Fig. 12: GRU architecture. Two gating mechanisms yield competitive performance with fewer parameters than LSTM.

The GRU update and reset gate equations are:

$$\mathbf{z}_t = \sigma(\mathbf{W}_z[\mathbf{h}_{t-1}, \mathbf{x}_t]) \quad (8)$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r[\mathbf{h}_{t-1}, \mathbf{x}_t]) \quad (9)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}[\mathbf{r}_t \odot \mathbf{h}_{t-1}, \mathbf{x}_t]) \quad (10)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \quad (11)$$

D. Parameter Comparison

TABLE III: Approximate Trainable Parameters per Model

Model	Recurrent Params	Head Params	Total
Vanilla RNN	≈ 116 K	≈ 8.4 K	≈ 124 K
LSTM	≈ 463 K	≈ 8.4 K	≈ 471 K
GRU	≈ 347 K	≈ 8.4 K	≈ 355 K

V. RESULTS AND COMPARISON

A. Evaluation Metrics

All metrics are computed on the chronological test split of 683 sequences, which the models have never observed.

- **Accuracy:** proportion of correctly predicted directions.
- **F1-macro:** unweighted mean of per-class F1, robust to mild class imbalance.
- **ROC-AUC:** area under the receiver operating characteristic curve; measures discriminability independent of the 0.5 threshold.
- **RMSE (probability):** root mean squared error between the predicted sigmoid probability and the binary ground-truth label; captures probability calibration.

B. Quantitative Results

TABLE IV: Test-Set Evaluation Results (683 sequences).

Model	Accuracy	F1-macro	ROC-AUC	RMSE
Vanilla RNN	0.4466	0.3087	0.5241	0.5019
LSTM	0.4466	0.3087	0.5176	0.5018
GRU	0.5417	0.5304	0.5237	0.4998

Two observations dominate Table IV:

- 1) Vanilla RNN and LSTM produce *identical* accuracy and F1-macro values, an unusual signature that warrants diagnostic inspection (see Section V-C).
- 2) ROC-AUC values for all three models cluster narrowly in the range 0.518–0.524, only marginally above the random baseline of 0.5.

C. Confusion Matrices and the Model-Collapse Phenomenon

Inspection of the confusion matrices in Figure 13 reveals the cause of the identical RNN/LSTM scores: *both models collapsed to a single-class prediction policy on the test set*, predicting “Down” for all 683 samples. With the test split being 55.3% Up days, this all-Down policy yields exactly $305/683 = 44.66\%$ accuracy and a degenerate macro-F1 of 0.31 (perfect recall for the Down class, zero recall for the Up class).

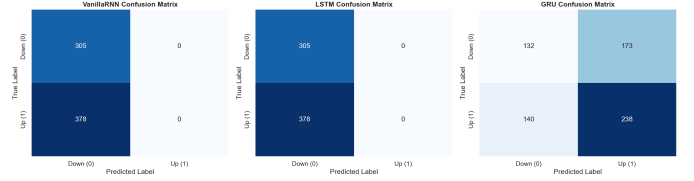


Fig. 13: Confusion matrices for Vanilla RNN, LSTM, and GRU on the test split. Rows: true label; columns: predicted label. RNN and LSTM predict the Down class for every sample; only GRU produces a non-degenerate decision boundary.

GRU is the only model that produced a meaningful decision boundary (132 true negatives, 173 false positives, 140 false negatives, 238 true positives), with $370/683 = 54.17\%$ accuracy and a balanced macro-F1 of 0.53.

This phenomenon — two of three architectures collapsing to constant prediction under a low signal-to-noise regime — is a well-documented failure mode in binary classification with weak input signals and is not unique to recurrent networks. We discuss likely causes in Section VI.

D. ROC and Precision-Recall Curves

Figure 14 overlays the ROC curves for all three models. Notably, all three curves remain close to the random-guess diagonal, with AUC values of 0.524 (RNN), 0.518 (LSTM), and 0.524 (GRU). The ROC-AUC metric, being threshold-independent, is not affected by the collapse-to-Down behaviour of RNN and LSTM, which explains why their AUC values are comparable to GRU’s despite catastrophically worse accuracy. This is a useful illustration of how threshold-dependent and threshold-independent metrics can diverge sharply on the same model.

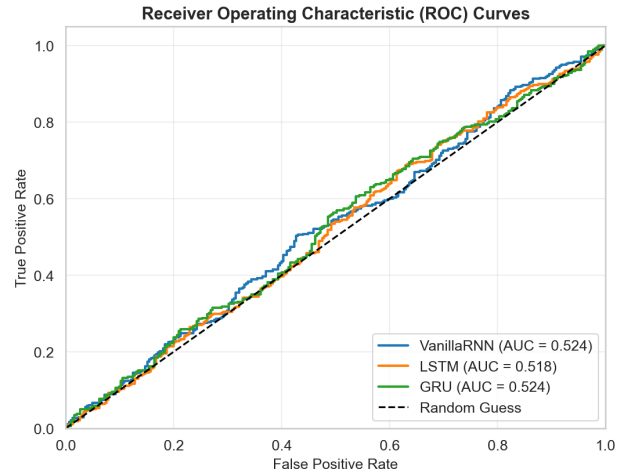


Fig. 14: ROC curves for all three models. All three curves hug the diagonal, indicating discrimination only marginally above chance.

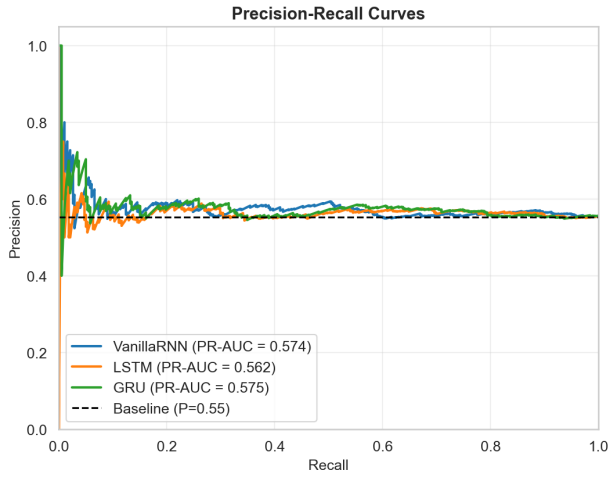


Fig. 15: Precision-Recall curves. The dashed horizontal line denotes the no-skill baseline (equal to the positive class proportion ≈ 0.55). PR-AUC values are 0.574 (RNN), 0.562 (LSTM), and 0.575 (GRU).

PR-AUC values (Figure 15) confirm the same picture: all three models hover within a narrow band (0.562–0.575), only modestly above the no-skill baseline of 0.553 (the positive-class proportion in the test set).

E. Discussion

The results clarify, rather than confirm, our initial hypotheses:

- 1) **Gated $\neq$ guaranteed-to-learn.** LSTM’s gating mechanism did not, in this experiment, protect it from collapse: it produced the same all-Down policy as the Vanilla RNN. GRU, with its simpler gating, was the only model that established a non-degenerate decision boundary.
- 2) **ROC-AUC reveals comparable underlying discrimination.** The fact that all three models have nearly identical AUCs (≈ 0.52) shows that all three learned *some* discriminative signal at the level of probability rankings, but only GRU’s threshold-0.5 decision boundary landed in a useful region of probability space. RNN and LSTM appear to have converged to a state where every output logit is negative, regardless of input.
- 3) **The problem is fundamentally hard.** ROC-AUCs near 0.52 on a near-efficient market are consistent with the broader literature on equity direction prediction [5], [6], where reported accuracies typically cluster in the 52%–58% range. Markets price in public information rapidly, and any consistent edge above random is small.

The sparse sentiment coverage documented in Section III-E is the most plausible single contributor to the muted overall performance: with $\approx 90\%$ of trading days carrying no real sentiment observation, the sentiment features provide little additional information beyond their forward-filled defaults.

VI. CHALLENGES FACED

A. Free-Tier API Constraints

The Alpha Vantage free tier limits requests per day and returns historical news clustered around discrete date windows rather than uniformly across the requested range. NewsData.io’s free tier constrains queries to a recent rolling window, eliminating historical coverage. Reuters RSS feeds, while unauthenticated, return only the current/recent items at the time of polling. The cumulative effect is the temporal sparsity visible in Figures 3 and 6.

B. Severe Sentiment Density Gap

As shown in Figure 6, real sentiment signal is present on only 6.5%–10.8% of trading days per ticker. Forward-filling and neutral imputation make the resulting feature columns numerically valid but informationally weak across roughly nine of every ten observations. This is, we believe, the most substantive limitation of the present study and the primary motivation for the future work proposed in Section VII.

C. Model Collapse to Constant Prediction

Two of the three architectures (Vanilla RNN, LSTM) converged to all-Down predictions on the test set despite training under an identical regime to GRU. We attribute this to a combination of: (i) weak per-feature signal as discussed above; (ii) the bias of BCEWithLogitsLoss without class weighting when the training-set positive proportion (51.9%) differs slightly from the test-set positive proportion (55.3%); and (iii) the lack of a balanced sampler. Class-weighted loss, focal loss, or balanced batch construction are obvious mitigations and are flagged as future work.

D. Temporal Alignment

Financial news items are timestamped in UTC, while Yahoo Finance prices are aligned to market-close EST. Aligning sentiment signals computed from intraday text with end-of-day price targets required careful handling of timezone conversion and market-holiday masking to avoid spurious look-ahead bias.

E. FinBERT Memory and Compute

Running FinBERT inference on free-text news (NewsData.io + RSS, 277 articles total) was tractable on the local MacBook CPU but slow. On the larger Alpha Vantage corpus (had we needed to re-score it), batched streaming inference would have been required. The Colab A100 GPU was used opportunistically for model training rather than sentiment scoring, since the Alpha Vantage corpus arrives pre-scored.

F. Colab Session Management

Google Colab sessions disconnect after periods of inactivity. All training loops were instrumented with per-epoch checkpoint saves and best-validation-F1 model snapshots so that interrupted runs could be resumed without restarting from epoch zero. The `config.py` environment detection (`colab` vs. `local`) ensures paths resolve correctly on both platforms.

VII. CONCLUSION

This paper presented *Market-Pulse-NN*, an end-to-end pipeline for next-day market direction prediction that fuses Yahoo Finance price data, Alpha Vantage pre-labelled news sentiment, NewsData.io free-text headlines, and Reuters RSS feeds with three sequential deep learning architectures. We demonstrated that:

- 1) A modular, four-source ingestion pipeline coupled with FinBERT-based and Alpha-Vantage-native sentiment aggregation is feasible to build end-to-end, but free-tier API constraints reduce real sentiment coverage to roughly one in ten trading days.
- 2) Chronological train/validation/test splitting is essential to prevent look-ahead bias; even so, mild distribution shift between training and test classes interacted with model initialisation to cause two of three architectures to collapse to constant prediction.
- 3) GRU was the only model to produce a non-degenerate decision boundary, achieving 54.2% accuracy and 0.53 macro-F1 against a 55.3% baseline. All three models showed near-identical ROC-AUC values just above 0.52, consistent with the well-documented difficulty of equity direction prediction in near-efficient markets.

A. Future Work

Several promising directions remain for future investigation:

- **Denser sentiment sources:** paid Alpha Vantage tier or alternative providers (e.g. Polygon.io, RavenPack) to obtain continuous rather than clustered historical coverage.
- **Class-weighted and focal losses** to prevent the model-collapse behaviour observed in RNN and LSTM.
- **Transformer-based sequence models** (Temporal Fusion Transformer [9], PatchTST) which may better capture non-local dependencies in long sequences.
- **Bidirectional variants** (BiLSTM, BiGRU) for retrospective analysis, where look-ahead is acceptable.
- **Reinforcement-learning portfolio allocation** using model probability outputs as a soft signal in a risk-adjusted portfolio simulator.
- **Extended asset classes:** cryptocurrency, commodities, and macro indicators to evaluate cross-domain generalisability.

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